

## TRAINING

# Electrical Safety and Work at Height

Practical basics of safely operating electrical installations, work permits and high-level (steeplejack) work

Training course for electrical personnel, steeplejacks and line supervisors



## After the course you can

- 01** — Recognise electrical and height hazards, assess risks and set safe work boundaries.
- 02** — Correctly use fall protection PPE, dielectric equipment and portable protective earths.
- 03** — Flawlessly perform technical and organisational measures to prepare the worksite and admit crews.
- 04** — Act instantly in incidents: free victims from current, rescue a suspended worker and give first aid.

**The course's main goal  
— zero injuries, no  
contact with current and  
prevention of falls from  
height.**

## Five course parts

- 01 — Electrical safety basics and effects of electric current.
- 02 — Technical safeguards and electrical protective gear.
- 03 — Safe work organisation, electrical safety groups and permits.
- 04 — Safety at height and in high-level (steeplejack) work.
- 05 — Combined work, suspension rescue and first aid.

### COURSE

5

practical modules of  
comprehensive safety

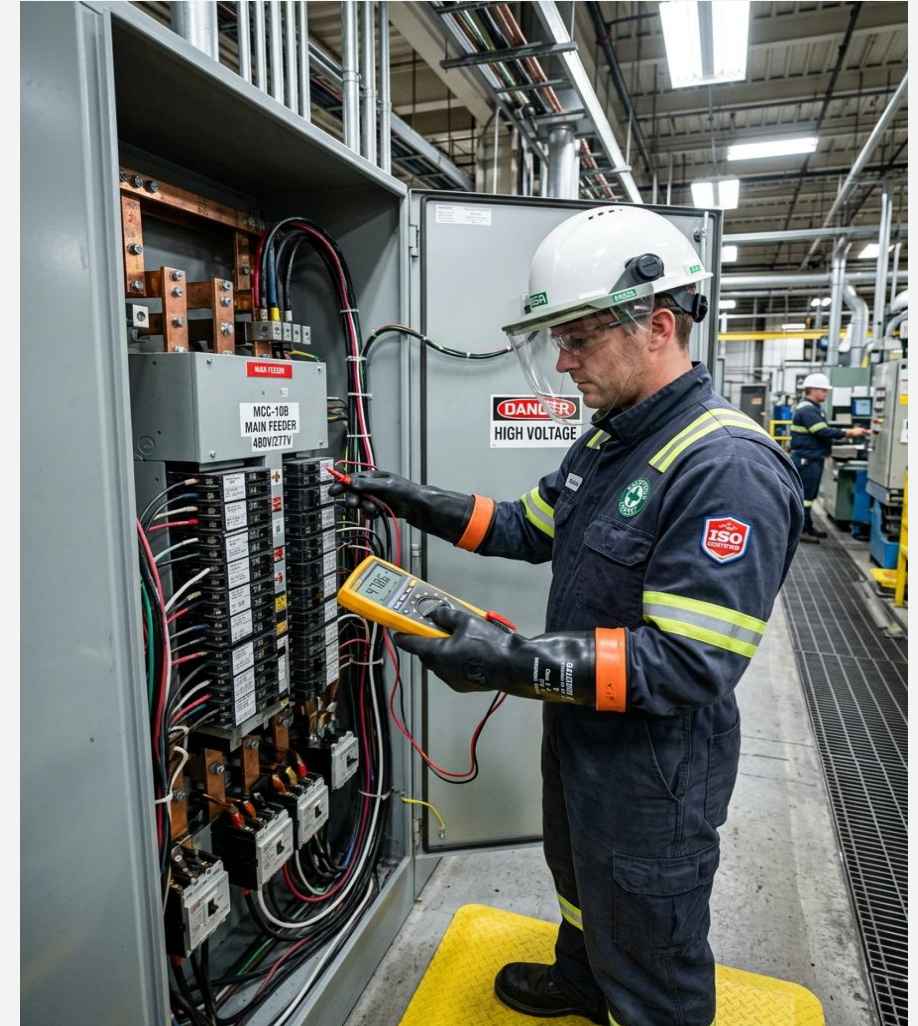


## Electrical safety basics

Electric current is invisible, odourless and acts instantly. The only protection is knowing the laws and strictly following the rules.

Reference: PUE RK • Safety Rules for the operation of electrical installations

Electrical safety (ES) and work at height • SafetyHUB





# Electrical safety at work

- **Work environment:** electrical installations up to and above 1000 V, switchboard rooms, cable routes and switchgear.
- **Nature of the hazard:** electric current shows no outward signs of danger (no colour, smell or sound in normal operation).
- **Human factor:** loss of vigilance, getting used to danger and ignoring the rules cause over 80% of severe electrical injuries.
- **Main rule:** every electrical installation is considered live until proven otherwise and until earthing is in place.

**Electric current forgives no mistakes: the lack of visible danger dulls your vigilance!**

## Effects of electric current on the human body

| TYPE OF EFFECT              | PATHOLOGICAL EFFECT                                              | CONTROL AND PROTECTION                                 |
|-----------------------------|------------------------------------------------------------------|--------------------------------------------------------|
| <b>Thermal</b>              | Severe tissue burns, vessel heating, boiling blood and lymph     | Insulation, protective shields, arc-rated PPE          |
| <b>Electrolytic</b>         | Decomposition of body fluids, blood breakdown, chemical changes  | De-energising, dielectric gloves and insulated tools   |
| <b>Mechanical (dynamic)</b> | Torn muscles, tendons and blood vessels, fractures, dislocations | Automatic disconnection (RCD, fast-acting relays)      |
| <b>Biological</b>           | Muscle spasms, cardiac fibrillation, respiratory arrest          | Equipotential bonding, TN neutral connection, earthing |

**Severity depends on the size of the current, the exposure time and its path through the body.**



## 4 thresholds of alternating current (50 Hz)

1

0.6 — 1.5 mA

Perceptible current: slight tingling, faint itch in the fingers.

2

10 — 15 mA

Non-let-go current: irresistible muscle spasms, the hand cannot unclench.

3

50 — 100 mA

Fibrillation current: chaotic heart contractions, circulatory arrest.

4

> 5 A

Lethal current: instant heart paralysis, the most severe thermal burns.

**A current of just 0.1 A (100 mA) for 1–2 seconds is fatal!**

## Local electrical injuries and effects

- **Electrical burns:** contact current burns (current passing through tissue) and arc burns (arc temperature reaches 4000–6000 °C).
- **Electrical marks:** firm grey or yellowish spots of dead skin with sharp edges at the current entry and exit points.
- **Skin metallisation:** tiny splashes of molten metal driven into the epidermis by an electric arc.
- **Electro-ophthalmia:** inflammation of the cornea and conjunctiva from intense arc UV radiation (goggles or a face shield required).

**Electric arc temperature reaches 6000 °C — working without a face shield and arc suit is deadly!**

## Current paths through the body (current loops)

- **Danger of a current loop:** set by the share of total current passing through vital organs — the heart, lungs and brain.
- **“Hand — hand” loop:** ~3.3% of the current passes through the heart. The most common workplace case when touching two phases.
- **“Left hand — feet” loop:** up to 4.0% of the current reaches the heart — the most critical, dangerous path!
- **“Foot — foot” loop:** step voltage current; ~0.4% passes through the heart, but falling to the ground turns it into “hands — feet”.

**A fall in the current zone instantly turns the loop from “foot-foot” to the deadly “hands-feet”!**



## CRITICAL BAN: Dangerous approach distances



- Do not approach unguarded live parts closer than the set minimum distances (up to 1 kV — 0.6 m; 1–35 kV — 0.6 m; 110 kV — 1.0 m; 220 kV — 2.0 m).
- Do not work alone in electrical installations where the work permit requires a second person to be present.
- Do not touch live insulators and structures without basic electrical protective equipment.
- Do not continue work if there are signs of arcing, an unusual transformer hum or a smell of burning.

**A person or tool coming closer than permitted causes flashover across the air gap!**

## Premises classified by electrical hazard

- **Premises without increased danger:** dry, normal temperature, insulating wooden floors, no conductive dust.
- **Premises with increased danger:** damp (humidity >75%), conductive dust, conductive floors (metal, concrete), temperature >35 °C.
- **Especially dangerous premises:** extreme damp (humidity near 100%), chemically aggressive environment or two signs of increased danger at once.
- **Outdoor switchgear areas:** always equated to the category of especially dangerous premises!

**Any work outdoors or in the open air is automatically treated as work in especially dangerous conditions!**



# Touch voltage and step voltage

- **Step voltage:** the potential difference between two points on the ground one human step apart (0.8 m).
- **Current spreading zone:** occurs when an OHL wire falls to the ground or on an earth fault. Danger zone radius — 8 metres (indoors 4–5 m).
- **Potential distribution:** falls hyperbolically from the fault point outwards; the largest difference is near the fallen wire.
- **No hopping on one foot:** losing balance and falling onto your hands is deadly, as it creates a “hands — feet” current loop.

**The danger radius  
around a fallen wire is 8  
metres!  
Never approach a wire  
on the ground!**



## 4 rules for leaving a step voltage zone

1

### Feet together

Immediately bring your feet together so there is no potential difference.

2

### “Shuffle”

Move slowly, with the heel of one foot against the toe of the other, never lifting.

3

### Step $\leq 10$ cm

Step length must not exceed the foot's width; no wide steps!

4

### Get 8 m away

Keep moving until fully out of the 8-metre danger radius around the fallen wire.

**Never run or jump — losing your balance will be fatal!**



## Section I summary: 5 electrical safety barriers

- **1. Hazard awareness:** remember that current is invisible and never be careless in any operation on electrical installations.
- **2. Distance barrier:** strictly keep the minimum permissible distances to live parts.
- **3. Environmental control:** account for humidity, conductive floors and open structures, which cut body resistance tenfold.
- **4. Behavioural rules:** no working alone where the rules require a crew or supervision.
- **5. Incident readiness:** solid knowledge of step voltage and the first aid algorithm for a victim.

**Electrical safety is not a set of tips but the law of survival on an industrial site!**

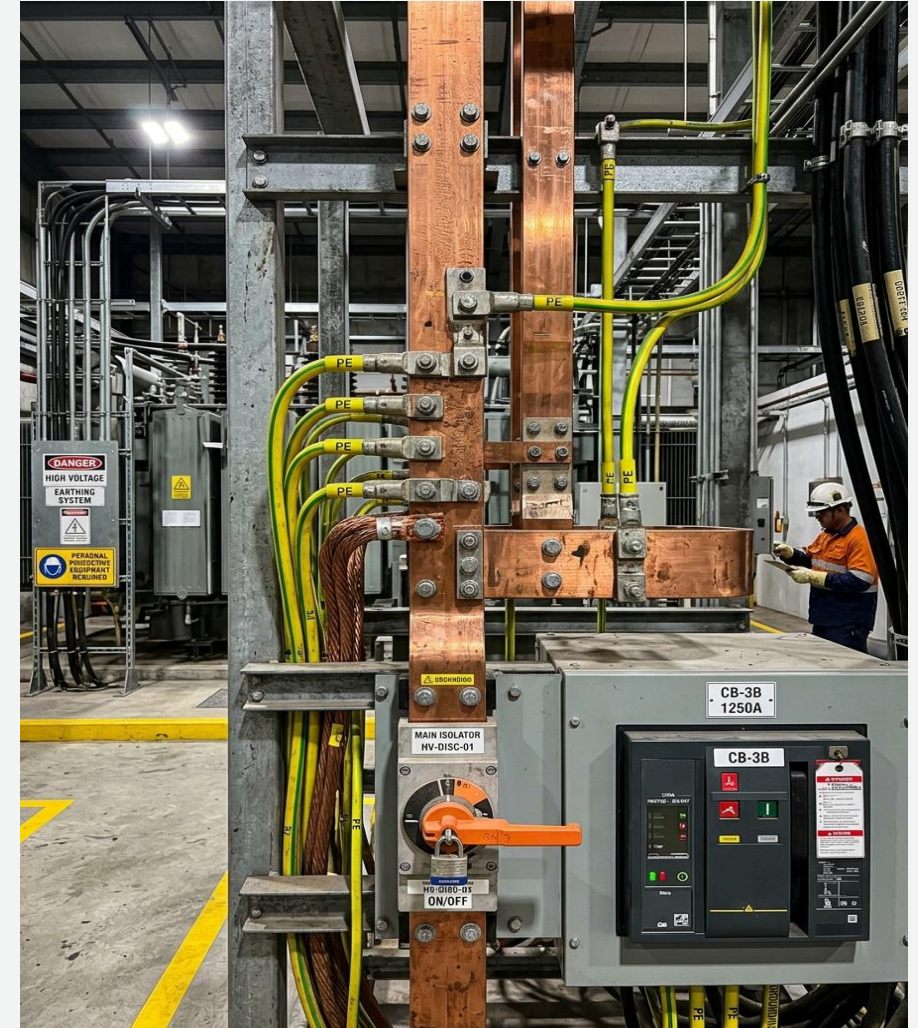


## Technical safeguards

Reliable earthing, insulation, extra-low voltage and electrical protective equipment form an impassable technical barrier.

Reference: PUE RK Chapter 1.7 • GOST 12.1.019-2017

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## Protective earthing: how it works

- **Definition:** a deliberate electrical connection of the non-current-carrying metal parts of an installation to an earthing device.
- **Purpose:** to reduce touch voltage to a safe level when phase conductor insulation breaks down to a metal enclosure.
- **Physical principle:** the earth electrode resistance ( $R_e \leq 4 \text{ Ohm}$ ) and the human body ( $R_b \sim 1000 \text{ Ohm}$ ) in parallel.
- **Result:** most of the fault current flows to earth through the low-resistance earthing loop, protecting the person.

**Current takes the path of least resistance: good earthing takes the hit!**

# Earthing systems of power networks

- **TN-S system:** the neutral (N) and protective (PE) conductors are separate throughout the network — the safest system.
- **TN-C-S system:** N and PE functions are combined in one PEN conductor in the upstream part and split at the building entry.
- **TN-C system:** an outdated system with a combined PEN throughout; prohibited for new and reconstructed installations!
- **TT and IT systems:** TT — independent earthing of the installation; IT — isolated neutral with insulation monitoring (mines, mobile units).

**Separating PE and N conductors is the key to correct RCD operation and electrical safety!**

## Protective neutral (TN): purpose and tripping

- **TN connection:** a deliberate electrical connection of exposed conductive parts to the solidly earthed neutral of the supply via the PE conductor.
- **Protection principle:** turns a phase-to-enclosure fault into a short circuit (SC) between the phase and the protective conductor.
- **Protection tripping:** the short-circuit overcurrent instantly trips the circuit breaker or blows the fuse link.
- **Standard time:** the faulty section of a 220/380 V network must be disconnected within 0.4 seconds (per the PUE).

**TN connection turns a fault into a short circuit and cuts the faulty section in a split second!**

## Residual current devices (RCD)

- **Differential transformer:** continuously compares the vector sum of currents entering the load via phase and returning via neutral.
- **Leakage current:** when a person touches a live part, a residual current (earth leakage) arises and induces an EMF in the secondary winding.
- **Tripping threshold:** to protect people against direct contact, RCDs with a rated residual current  $I_{\Delta n} \leq 30 \text{ mA}$  are used.
- **Response speed:** a quality RCD disconnects the faulty circuit in 0.02–0.04 seconds, before heart fibrillation sets in.

**A 30 mA RCD is the only protection that can save a life in direct contact with a bare wire!**



## Extra-low voltage (ELV / SELV)

- **Definition:** rated voltage not exceeding 50 V AC or 120 V DC between conductors or to earth.
- **Application:** supply of portable hand-held power tools and portable lamps in conditions of increased danger.
- **Voltage levels:** 42 V (or 36 V) — in premises with increased danger; 12 V — in especially dangerous conditions (inside boilers, tanks, wells).
- **Power source:** only via isolating step-down transformers with separate windings (autotransformers are prohibited!).

**Inside metal tanks and wells, only lamps rated at no more than 12 volts may be used!**

## Electrical protective equipment types

- **Basic up to 1000 V:** dielectric gloves, tools with insulated handles, voltage indicators, clamp meters.
- **Additional up to 1000 V:** dielectric overshoes, dielectric mats, insulating stands and covers, protective barriers.
- **Basic above 1000 V:** insulating rods of all types, insulating tongs, voltage indicators above 1000 V.
- **Additional above 1000 V:** dielectric boots, dielectric gloves (above 1 kV they are only additional!), mats.



**Basic PPE withstands the operating voltage; additional PPE is used only together with basic PPE!**



## Electrical protective equipment: 4 checks

1

### Test stamp

Check the next test date on the stamp.  
Expired PPE is prohibited!

2

### Visual check

Make sure there are no cracks, punctures, tears, chips or mechanical defects.

3

### Roll test

Roll the glove towards the fingers to check for punctures under air pressure.

4

### Clean and dry

Before work, wipe off dust, dirt, oil and moisture with a clean, dry cloth.

**Using electrical protective equipment with an expired test stamp is strictly prohibited!**



## 4 steps to verify absence of voltage

1

### Self-check

Check that the indicator works on parts known to be live or with a test device.

2

### Phase by phase

Verify there is no voltage between each phase and earth, then between all phases.

3

### Induced voltage

Make sure there is no dangerous induced voltage from adjacent energised lines and phases.

4

### Re-test

Re-check the indicator on a known live source to rule out a sudden failure.

**Always wear dielectric gloves when verifying absence of voltage!**



## Portable protective earthing

- **Purpose:** protect workers from voltage applied in error by switching devices and from induced voltage.
- **Fitting order:** FIRST connect the earth clamp to the earthing device (loop), and only THEN to the live parts.
- **Removal order:** strictly in reverse: FIRST take the clamps off the live parts, and only THEN detach the clamp from the earth electrode.
- **Earthing conductor cross-section:** flexible copper wires of at least 16 mm<sup>2</sup> (up to 1000 V) and at least 25 mm<sup>2</sup> (above 1000 V) for thermal withstand.

**Fitting earths: first to Earth, then to Phase!**  
**Removing: first from Phase, then Earth!**  
**Wrong order is deadly!**

## Electrical safety signs and posters

- **Prohibition:** “DO NOT SWITCH ON! PEOPLE AT WORK”, “DO NOT OPEN! PEOPLE AT WORK” — hung on the drives of switching devices.
- **Warning:** “STOP! HIGH VOLTAGE”, “DO NOT CLIMB! DANGER OF DEATH”, “CAUTION! ELECTRICAL HAZARD” — mark off the danger zone.
- **Mandatory:** “WORK HERE”, “CLIMB HERE” — clearly mark the prepared and verified safe workplace.
- **Indicative:** “EARTHED” — hung on device keys and control handles and on earthed sections of the installation.

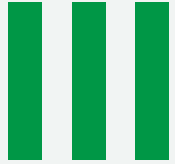
**The “DO NOT SWITCH ON” sign has the legal force of an order!**  
**Only the person who hung it may remove it!**

## CRITICAL BAN: Breaching technical measures



- Do not start work in an electrical installation without completing all 5 mandatory technical measures to prepare the site.
- Do not switch on disconnected devices or remove prohibition signs and mechanical locks without authorisation.
- Do not use random wires of non-standard cross-section or twisted wires as portable protective earths.
- Do not touch live parts without verifying absence of voltage with a working indicator while wearing dielectric gloves.

**De-energising without earthing does not guarantee safety — induced voltage kills just like operating voltage!**



# Work organisation and permits

Clearly divided responsibility, strict admission rules and the work permit rule out the human factor and errors.

Reference: Safety Rules of the RK • Safety Rules • Admission to work

Electrical safety (ES) and work at height • SafetyHUB



## Two ways to authorise electrical work



### WORK PERMIT

- A written order on a prescribed form.
- Sets the exact place and the start and end times of work.
- Records the crew, responsible persons and safety measures.
- Required for de-energised work and work near live parts.
- Maximum validity — no more than 15 calendar days.



### WORK ORDER

- A verbal task for a one-off simple job.
- Valid for one work shift of the crew.
- Must be recorded in the site operational log.
- Done by a small team (1–2 people) with a clear scope.
- Used for unplanned urgent minor work up to 1 kV.

**Routine operational work follows a pre-approved list, without a work permit or work order.**

## Electrical safety qualification groups

- **Group I:** non-electrical staff (office, warehouses, PC operators); briefing once a year, recorded in the log.
- **Group II:** electrotechnological staff (welders, crane operators, machine operators); 72+ hours of training, knows installations up to 1 kV.
- **Group III:** electrical staff; 1–3 months' experience in Group II, may inspect alone and work up to 1 kV.
- **Group IV:** may work in installations above 1 kV and act as work supervisor, permit issuer or person in charge of electrical facilities up to 1 kV.
- **Group V:** highest level (engineers, energy managers, persons in charge of installations above 1 kV); knows all circuits and rules.

**A safety group is a personal measure of responsibility and the right to independent access to hazardous work!**

## Persons responsible for safe work

- **Permit issuer (work order giver):** decides if work is needed, its safe scope and the qualifications of the crew's responsible persons.
- **Responsible work manager (RWM):** ensures the planned safety measures are complete and sufficient and coordinates the crews.
- **Permitting officer:** responsible for correct workplace preparation, checking absence of voltage, fitting earths and signs.
- **Work supervisor (WS):** responsible for a clear on-site crew briefing, correct use of PPE and continuous supervision.
- **Observer and crew members:** the observer supervises non-electrical staff; crew members strictly follow the WS's instructions.

**Each responsible person answers for their part of preparation — combining roles is strictly regulated by the SR!**



## 6 steps to prepare the workplace

- **1. Disconnection:** make the required disconnections with a visible break in the circuit.
- **2. Locks and signs:** lock the drives with LOTO locks and hang prohibition signs.
- **3. Voltage check:** verify absence of voltage on all phases with a working indicator.
- **4. Earthing:** close the fixed earthing switches or apply portable earths.
- **5. Indicative signs:** hang “EARTHED” signs on device keys and handles.
- **6. Fencing:** fence off the workplace with portable screens and hang safety signs.

**Strict order of the 6 preparation steps:**  
**disconnect → lock → sign → test → earth → fence!**

# LOTO protective lockout in power networks

- **LOTO concept (Lockout / Tagout):** mechanical isolation of hazardous energy sources with personal padlocks and warning tags.
- **Personal lock:** each crew member puts their own lock on the breaker or switch lockout device and keeps the only key.
- **Group lockout:** with several crews, a group lock box is used: equipment keys go inside, and every team puts its lock on the box.
- **Error-proofing:** no breaker can be switched on remotely or manually until the last worker removes their personal lock.

**One worker — one lock  
— one key!**  
**No one can energise the  
line while you are  
working on it!**



# 4 steps to admit a crew under a work permit

1

### Staff check

Check the crew by names, ID certificates and electrical safety groups.

2

### Show on site

Prove absence of voltage, show the earths, boundaries and adjacent equipment.

3

### Task briefing

Explain the work method, danger zones, escape routes and safe work practices.

4

### Sign the permit

Permitting officer, work supervisor and crew sign the permit in person.

**No work may start without the permitting officer's and work supervisor's signatures on the permit!**

## Continuous supervision during work

- **Duty of the work supervisor (observer):** continuously supervise all crew members throughout the task.
- **No leaving:** the work supervisor must not combine supervision with any other work or leave the workplace.
- **Supervisor's temporary absence:** if the work supervisor has to leave, the entire crew must be withdrawn from the installation.
- **Crew members:** may not stay in the switchgear or at height without the work supervisor; switchgear doors are locked during breaks.

**The work supervisor is the crew's “safety eyes”:  
doing their own work  
means losing control over  
people!**

## Breaks and completion of work

- **Lunch break:** the crew leaves the work zone; the permit stays with the work supervisor; signs, earths and barriers stay in place.
- **Re-admission after a break:** the work supervisor personally checks that the workplace is unchanged and admits the crew without the permitting officer.
- **Daily end of work:** the workplace is tidied up and the permit is handed to the duty operating staff, marked as end of shift.
- **Final completion:** temporary fixtures are dismantled, crew earths removed, and the permit is closed with WS and permitting officer signatures.

**After the shift, switching on the installation before the work permit is formally closed is strictly prohibited!**



## 4 work supervisor actions during a break

1

### Withdraw crew

Stop the tasks, collect all tools and lead people out of the electrical installation.

2

### Lock access

Lock the switchroom doors, barriers and gates of the work zone.

3

### Keep the setup

Leave all signs, LOTO locks and portable protective earths in place.

4

### Check before re-entry

Before resuming, personally confirm the setup is safe and the earths are intact.

**Crew members must not enter the work zone before the work supervisor returns!**

## Admission of seconded staff and contractors

- **Secondment letter:** the employer sends an official letter listing the workers, their safety groups and the rights of responsible persons.
- **Knowledge check:** seconded staff receive induction and initial briefings on the client's electrical installation layout.
- **Admission certificate:** issued when a separate fenced work zone is allocated to the contractor on an operating site.
- **Assigning the permitting officer:** initial admission of seconded crews is always done only by the client's own operating staff.

**A contractor on site calls for extra attention: initial admission is done only by the owner company's own staff!**

## Section III summary: Organisational control chain

- **Unbroken chain:** safety is guaranteed only when every link works — permit, briefing, LOTO and supervision.
- **Personal responsibility:** each worker signs the work permit for their own and their colleagues' safety.
- **Documenting changes:** any change in the crew or work conditions requires the admission to be reissued at once.
- **Life comes first:** organisational rules are based on the analysis of hundreds of real incidents and are mandatory.

**A work permit is not a formality but a legal document that protects your life and freedom!**



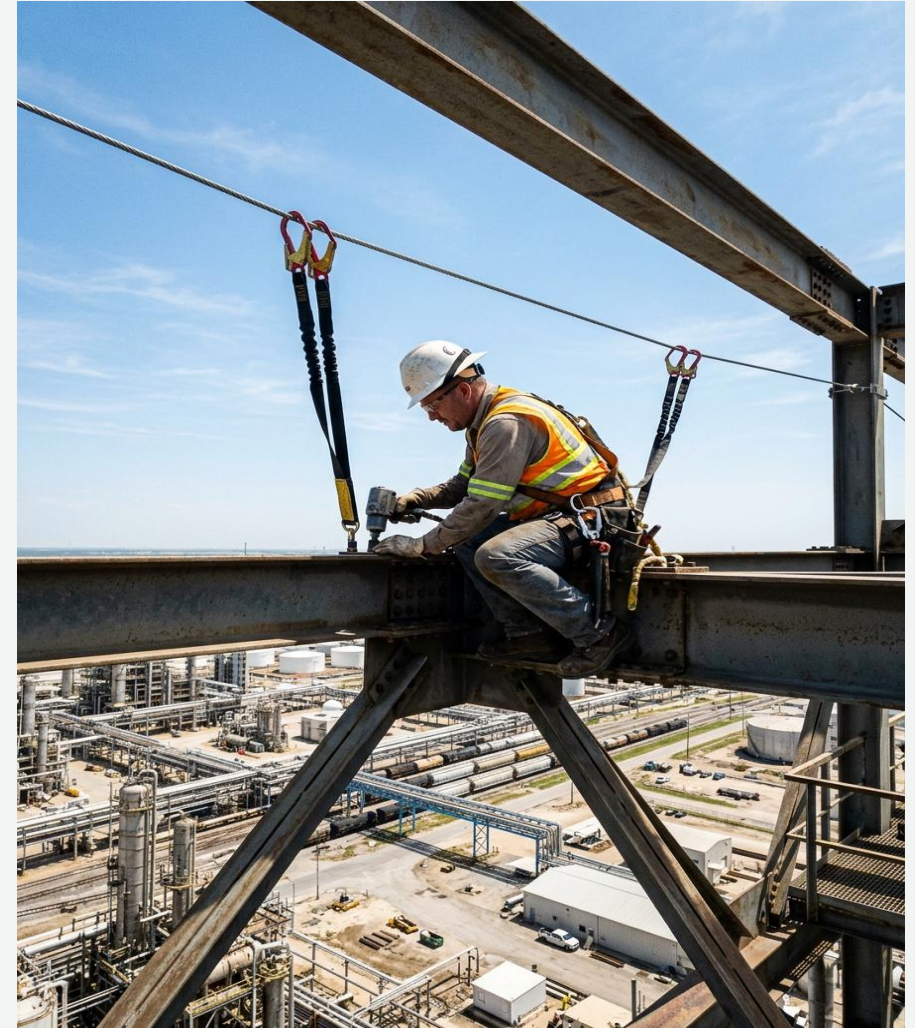
# IV

## Work at height and steepjack work

Falls from height are the leading cause of fatal workplace injuries. The right choice of fall protection saves lives.

Reference: Rules on occupational safety for work at height

Electrical safety (ES) and work at height • SafetyHUB



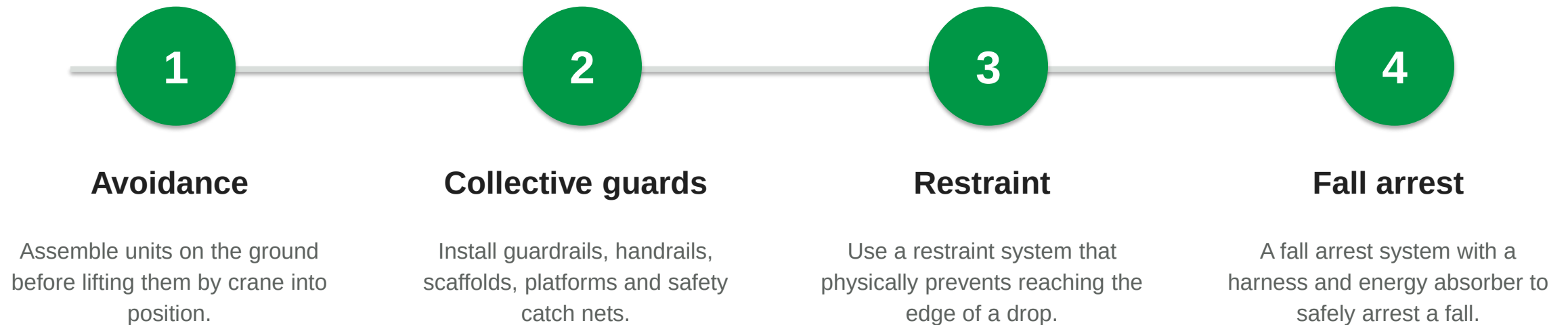


## Work at height defined: regulatory criteria

- **Basic height criterion:** work with a risk of falling from 1.3 m (under construction codes) or 1.8 m or more above the surface.
- **Guarding:** work on platforms within 2 m of unguarded drops over 1.3 m, or where guardrails are lower than 1.1 m.
- **Hazards below:** work less than 1.3 m above machines, mechanisms, protruding objects, water or molten metal.
- **High-level (steeplejack) work:** work more than 5 m above the ground or a floor slab, done from erection fixtures or directly from structures.

**Even an unprotected fall from just 1.8 metres in 85% of cases leads to disability or death!**

## Fall protection hierarchy: 4 levels



**Fall protection PPE is the last line of defence! Collective guards always come first!**

## Safety groups for work at height

- **No group:** work from scaffolds, platforms and towers with guardrails of 1.1 m or more, and from leaning ladders up to 5 m.
- **Group 1:** workers admitted to work as part of a crew or under the direct control of an appointed supervisor.
- **Group 2:** foremen, crew leaders, responsible performers and safety specialists who carry out periodic PPE inspections.
- **Group 3:** responsible work managers under permits, persons approving the WEP for work at height, certification board members, rescuers.

**A Group 1 worker may not work alone — only under the direct control of a Group 2 or 3 specialist!**

## Two protection systems: Restraint vs Fall arrest



### RESTRAINT SYSTEM

- Goal: physically PREVENT access to the edge of a drop.
- Components: a work belt or harness + a fixed-length lanyard.
- No fall: the fall factor is zero (no dynamic shock load).
- Load: no energy absorber needed, as a fall is impossible.
- Limitation: never use where a fall into a void is possible!



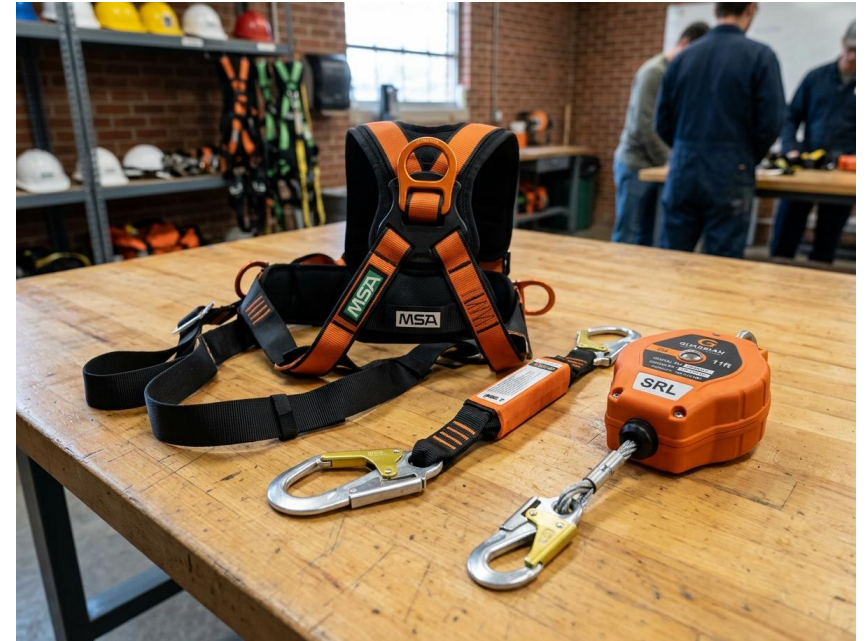
### FALL ARREST SYSTEM

- Goal: SAFELY ARREST a fall that has already begun.
- Components: full body harness + energy absorber + anchor line.
- Shock absorption: the absorber cuts the arrest force to  $\leq 6$  kN.
- Load distribution: the load is transferred to the thighs, pelvis and chest.
- Prerequisite: calculate the clearance below the worker.

**A work positioning belt (without shoulder and thigh straps) is PROHIBITED as a fall arrest system!**

## Fall arrest system components: The ABC rule

- **A (Anchor):** a reliable anchor device (sling, clamp, beam clamp) with a strength of at least 22 kN (2.2 tonnes).
- **B (Body wear):** a full body harness with shoulder and thigh straps and a dorsal fall arrest D-ring.
- **C (Connector):** a lanyard with an energy absorber, a self-retracting lifeline (SRL) or a rope grab on a flexible line.
- **100% tie-off rule:** when moving along structures, use a twin-leg (Y-shaped) lanyard — one leg is always clipped in!



Fall protection PPE works only with correct anchorage and a calculated clearance!



## Fall factor and clearance

- Fall factor = Fall height / Lanyard length. It determines how hard the dynamic shock on the worker's body is in a fall.
- **Factor 0 (anchor above head):** the best option, minimal fall distance, gentle absorber deployment.
- **Factor 1 (anchor at chest/shoulder level):** a standard fall of the slack lanyard length; acceptable with an energy absorber.
- **Factor 2 (anchor at foot level):** CRITICAL risk! Free fall equals twice the lanyard length (up to 4 m); risk of hitting structures.
- **Clearance:** the calculated distance below the workplace (lanyard length + absorber extension 1.2 m + height 1.8 m + margin 1.0 m = 5.7–6.0 m).

**Clip the lanyard AS HIGH  
ABOVE YOUR HEAD AS  
POSSIBLE!**

**Factor 2 — extreme load  
on the anchor and spine!**

## Anchor devices and rigid / flexible lines

- **Class A:** fixed structural anchor points on concrete and steel load-bearing building structures.
- **Class B:** portable temporary anchor devices (textile slings, tripods, I-beam clamps).
- **Class C:** flexible horizontal cable anchor lines for free, continuous movement of the worker along the structure.
- **Class D:** rigid horizontal rail anchor lines with a sliding trolley for heavy-duty use.
- **Never anchor to unreliable supports:** process pipelines, cable trays, radiators and window frames!

**Never attach the lanyard to process pipes or cable trays — only to verified load-bearing structures!**



## Ladders, stepladders and access platforms

- **Three points of contact rule:** when climbing up or down a ladder, always keep two hands and one foot (or two feet and one hand) on it.
- **Leaning ladder angle:** strictly 70–75 degrees to the horizontal (height-to-base ratio 4:1).
- **Top end:** a leaning ladder must extend at least 1.0 metre above the landing.
- **Work ban:** do not work from the top two steps of stepladders without handrails and guarded platforms.
- **No heavy work:** heavy erection and power electrical installation work must not be done from leaning ladders.

**A leaning ladder is a means of access, not a workplace!**  
**Heavy work is done only from scaffolds and platforms!**



# 4 steps of a pre-use harness check

1

### Webbing

Check the webbing for cuts, burns, abrasion and intact contrast stitching.

2

### Metal hardware

Inspect D-rings and buckles for deformation, cracks, corrosion and secure latches.

3

### Fall indicator

Make sure the fabric fall indicator is intact (the harness has not arrested a fall).

4

### Body fit

Adjust the thigh and shoulder straps: a flat hand should fit snugly under the thigh strap.

**A harness that has arrested a fall is immediately withdrawn from use and destroyed!**

Reference: Rules for periodic inspection and rejection of fall PPE



## Risk matrix for erection work at height

| RISK / SOURCE                            | CONSEQUENCE                                                      | CONTROL AND PROTECTION                                        |
|------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Person falling from height</b>        | Severe multiple trauma, spinal fractures, death                  | 100% tie-off, Y-lanyard, horizontal anchor line               |
| <b>Falling objects and tools</b>         | Injury to personnel at ground level                              | Helmets, protective canopies, tool lanyards, 15 cm toe boards |
| <b>Scaffold/platform instability</b>     | Collapse of the structure with people and materials              | Anchoring to the facade, diagonal bracing, formal acceptance  |
| <b>Weather (wind, thunderstorm, ice)</b> | Blown off by a gust, loss of balance, slipping, lightning strike | Monitor forecasts, stop work at once if wind >10–12 m/s       |

**The danger zone below work at height must be fenced off with warning tape and signs!**

## CRITICAL BAN: Violations during work at height



- Being at height without a correctly worn and adjusted full body harness and a locked carabiner.
- Wind speed above 10–12 m/s (when erecting solid structures and panels) or 15 m/s for ordinary work.
- Onset of a thunderstorm, squall, fog (visibility under 50 m) or icing of structures and ladders.
- No approved work execution plan for work at height (WEP) or rescue and evacuation plan (REP).
- Using homemade, uncertified or damaged platforms, barrels or stands instead of scaffolding.

**An unprotected fall gives no second chance — safety at height is absolute!**

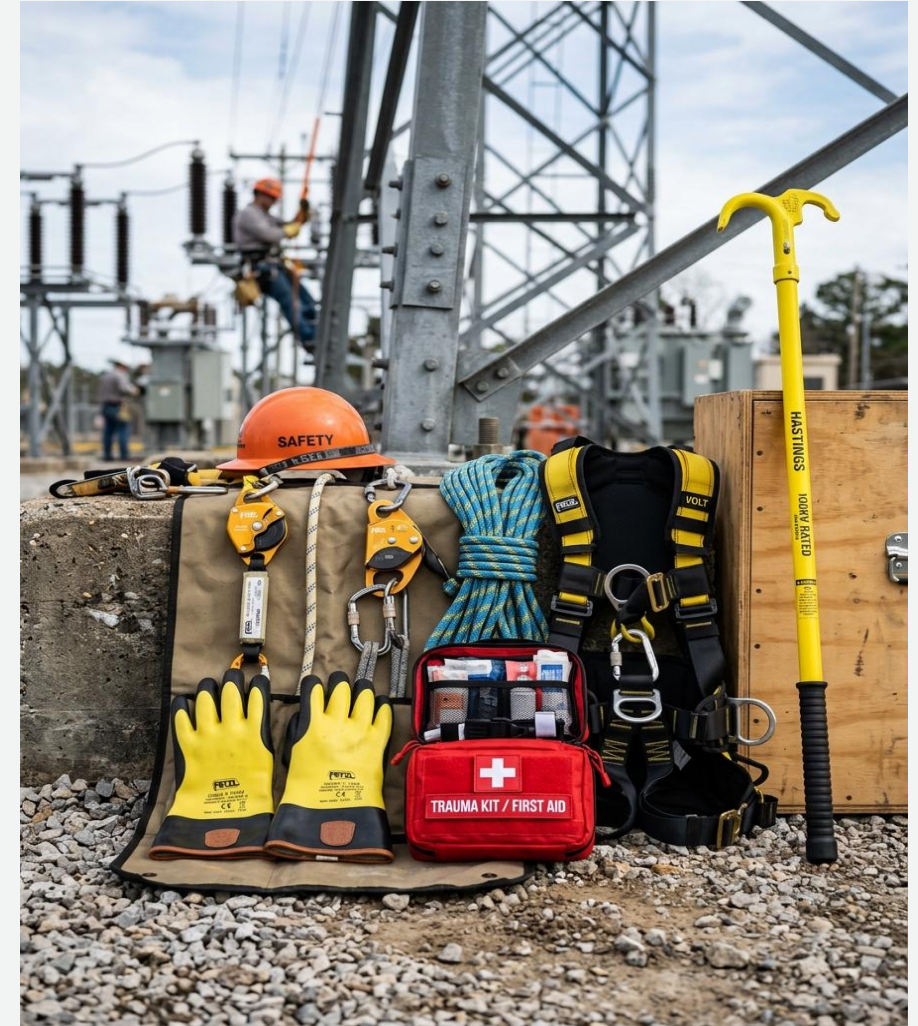
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## Combined work, rescue and first aid

Specifics of electrical work at height, preventing suspension trauma, emergency evacuation and first aid for victims.

Reference: REP • Cross-industry first aid instruction

Electrical safety (ES) and work at height • SafetyHUB





# 4 safety steps for work on OHL poles

1

### Pole check

Check for wood rot or cracks in reinforced concrete, pole stability and guy wires.

2

### 100% tie-off

Use a flexible anchor line or a pole strap throughout the climb.

3

### Safety zone

Keep safe clearances to adjacent lines that are live or carry induced voltage.

4

### Earth the line

Fit portable earths on both sides of the worksite on all phase conductors.

**Climbing angle poles from the inside of the conductor angle is strictly prohibited!**



# Aerial work platforms (AWP) near OHL

- **Work permit:** work within 30 metres of the outermost OHL conductor is done only under a work permit and a manager's supervision.
- **Chassis earthing:** the AWP vehicle body must be earthed with a flexible copper conductor to a fixed earth electrode or the substation earth.
- **100% tie-off:** an electrician in the basket must be clipped by the harness lanyard to the basket's dedicated certified anchor point.
- **No travel:** moving the AWP with people in the basket or with the boom raised is strictly prohibited!

**Earthing the AWP body and clipping into the basket are mandatory before switching on the boom hydraulics!**



## 4 stages of the rescue plan (REP)

1

### Alarm and timer

Alert the supervisor at once and start timing.  
Rescue time — up to 10 minutes!

2

### Rig rescue kit

Attach the rescue device to a secure anchor above the victim or on an adjacent structure.

3

### Hook and lower

Clip the rescue line to the suspended worker's harness, cut their lanyard, lower gently.

4

### Shock control

Place them in the right position (sitting or semi-sitting!), call an ambulance.

**The rescue and evacuation plan (REP) and rescue kit must be on site before work starts!**

## Suspension trauma in a harness

- **Injury mechanism:** thigh straps compress the femoral veins and blood pools in the legs, causing acute venous congestion.
- **Reduced blood return:** after 5–10 minutes of motionless hanging, brain hypoxia, loss of consciousness and cardiac arrest follow.
- **Self-help while suspended:** if conscious, actively move the legs (“pedal a bicycle”) and stand in the foot-loop stirrups.
- **GOLDEN RULE:** after lowering, NEVER lay the victim flat on their back! (risk of “reflow syndrome” and cardiac arrest).

**After lowering the victim,  
NEVER, EVER LAY THEM  
ON THEIR BACK!  
Keep them semi-sitting for  
30+ minutes to spare the  
heart from toxic shock!**

Reference: Medical emergency protocol for suspension trauma



# Releasing a victim from current at height

- **Step 1. De-energise the installation:** switch off the nearest isolator, breaker or switch with the emergency stop.
- **Step 2. Prevent a fall:** remember that once the current is off, the victim will go limp and fall unless secured by fall protection!
- **Step 3. Insulate the rescuer:** if power cannot be cut quickly, use dielectric gloves, an insulating rod or an axe with a dry handle.
- **Step 4. Separate the victim:** pull the victim by dry clothing only, with one hand, without touching bare skin or metal.

**When cutting the current to a victim at height, secure them from falling: the fall after a shock is deadlier than the current!**



## First aid: cardiopulmonary resuscitation (CPR)

- **Assessment:** check consciousness (call out, shake the shoulders) and breathing (the “listen — look — feel” rule for 10 seconds).
- **Chest compressions:** 30 presses on the centre of the breastbone, 5–6 cm deep, at 100–120 per minute with the heel of the hand.
- **Rescue breaths:** 2 mouth-to-mouth breaths through a barrier face shield after tilting the head back and pinching the nose.
- **Using an AED (defibrillator):** apply the automated defibrillator pads and follow the device's voice prompts.
- **Fractures and bleeding:** apply a pressure bandage to bleeding, splint fractures, do not reset bone fragments!

**Start CPR immediately!**  
**Every minute of delay cuts**  
**the brain's chance of**  
**survival by 10%!**  
**Ratio: 30 to 2!**

## CRITICAL STOP: 6 grounds for an absolute stop



- 1. Sudden appearance of voltage on disconnected live parts or sparking at the worksite.
- 2. Worsening weather: onset of a thunderstorm, squall wind (>12 m/s), snowstorm or dense fog.
- 3. Detection of a defect or tear in the harness, lanyard, carabiner or anchor line during work.
- 4. Sudden illness of any crew member (dizziness, nausea, weakness, chest pain).
- 5. Forced departure of the work supervisor or the person providing continuous supervision.
- 6. An emergency, or tools or structures falling near people.

**The right to stop work (“STOP CARD”) is every worker's sacred right, with no risk of punishment!**



## Final admission checklist: 6 safety conditions

- ✓ **Work permit issued** — Permit signed by all parties, still valid, task briefing given.
- ✓ **De-energised and earthed** — Switching devices off, absence of voltage verified, earths applied.
- ✓ **LOTO lockout in place** — Personal mechanical locks on the drives, keys held by crew members.
- ✓ **Electrical PPE checked** — Dielectric gloves, tools and indicators tested, stamps valid.
- ✓ **Full body harness fitted** — 100% continuous tie-off, anchor point reliable ( $\geq 22$  kN), twin-leg lanyard connected.
- ✓ **Rescue plan ready** — Evacuation kit on site, communications working, danger zone below fenced off.